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E-Commerce Sales Data Analysis Using Hadoop, MapReduce and Pig

Big Data Analytics — Project Report

Submitted To

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1. Introduction

Modern e-commerce platforms generate massive volumes of transactional data every day, capturing details about customers, products, order quantities, prices, payment methods, and delivery locations. Analysing this data manually becomes impractical as its volume grows, which creates a need for distributed Big Data technologies capable of storing and processing large datasets efficiently.

This project analyses a large e-commerce sales dataset using three core Big Data technologies:

- Hadoop Distributed File System (HDFS) for distributed data storage.
- Hadoop MapReduce for distributed data processing.
- Apache Pig for high-level data analysis using Pig Latin.

The complete workflow illustrates how a raw transactional e-commerce dataset can be stored on a distributed file system, processed using a MapReduce program, and further analysed using Apache Pig scripts to extract meaningful business insights.

2. Objectives

The main objectives of this project are:

1. To store a large e-commerce sales dataset using HDFS.
2. To process the dataset using Hadoop MapReduce.
3. To calculate city-wise total quantity sold and total revenue using MapReduce.
4. To calculate payment-method-wise order counts using MapReduce.
5. To analyse the dataset using Apache Pig.
6. To calculate category-wise average unit price, quantity, and revenue using Pig.
7. To calculate city-wise average order revenue using Pig.
8. To calculate payment-method-wise total revenue using Pig.
9. To demonstrate a complete, practical Big Data analytics workflow.

3. Dataset

The dataset used in this project is a large-scale e-commerce sales dataset consisting of 600,000 transactional records. Each record represents a single product order, including customer, product, category, quantity, pricing, revenue, and payment information across twenty different cities/districts.

3.1 Dataset Attributes

Attribute	Description
order_id	Unique identifier for each order
order_date	Date on which the order was placed
customer_id	Unique identifier of the customer
customer_name	Name of the customer
city	City/district where the order originated
product_id	Unique identifier of the product
product_name	Name of the product
category	Product category (e.g. Electronics, Grocery, Books)
quantity	Number of units purchased in the order
unit_price	Price of a single unit of the product
revenue	Total revenue generated by the order (quantity × unit_price)
payment_method	Mode of payment used (Cash, Card, Bank Transfer, Mobile Banking)

The dataset contains 10 product categories (Toys, Books, Beauty, Sports, Grocery, Clothing, Furniture, Automotive, Electronics, and Home Appliances), 20 cities, and 4 payment methods. The revenue field was used for all revenue-related calculations.

3.2 Dataset Size

The dataset file (sales_data.csv) is approximately 56.14 MB in size and contains 600,000 transaction records, which were stored in HDFS with a replication factor of 3.

4. Tools and Technologies

Technology	Purpose
Hadoop	Big Data processing framework
HDFS	Distributed data storage
MapReduce	Distributed data processing
Apache Pig	High-level data analysis using Pig Latin
Java	MapReduce programming language
Windows CMD	Command-line execution environment

5. Project Workflow

The overall project workflow consists of three major stages:

1. Store the dataset in HDFS.
2. Process the dataset using Hadoop MapReduce.
3. Analyse the dataset using Apache Pig.

Sales Dataset → HDFS → MapReduce → Pig Analysis → Results

6. HDFS Data Storage

Hadoop Distributed File System (HDFS) was used to store the e-commerce sales dataset (sales_data.csv). The dataset was uploaded to the following HDFS directory:

```
/ecommerce/inputs/sales_data.csv
```

The Hadoop NameNode web interface (localhost:9870) was used to verify that the dataset was successfully stored in HDFS, as shown below. The file is 56.14 MB in size with a replication factor of 3.

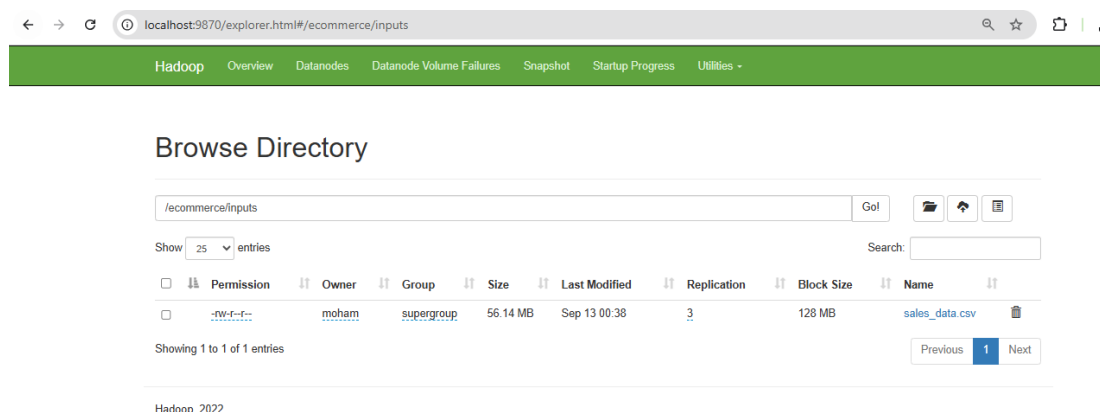


Figure 1: Verifying dataset storage in HDFS using the NameNode web UI

7. MapReduce Data Processing

Hadoop MapReduce was used to process the e-commerce dataset and compute city-wise sales statistics and payment-method-wise order counts.

The MapReduce program consists of two main components:

- Mapper: Reads each transaction record and emits a key (city or payment method) along with the associated quantity and revenue as values.
- Reducer: Groups the records by key and aggregates the total quantity and total revenue for each group.

The output was viewed directly from HDFS using the `hdfs dfs -cat` command, as shown below. The first result set shows city-wise total quantity and total revenue, and the second shows payment-method-wise order counts.

```
C:\hadoop\sbin>hdfs dfs -cat /ecommerce/mr_output_city/part-r-00000
Barisal Total Quantity = 151714.0, Total Revenue = 152593810.09
Bogra Total Quantity = 151634.0, Total Revenue = 152427937.81
Chattogram Total Quantity = 150153.0, Total Revenue = 150417629.19
Comilla Total Quantity = 152184.0, Total Revenue = 153011001.50
Coxs Bazar Total Quantity = 150078.0, Total Revenue = 151138595.19
Dhaka Total Quantity = 149769.0, Total Revenue = 150574011.48
Dinajpur Total Quantity = 149867.0, Total Revenue = 150411309.29
Faridpur Total Quantity = 149428.0, Total Revenue = 150301138.52
Feni Total Quantity = 148799.0, Total Revenue = 149524743.00
Gazipur Total Quantity = 150762.0, Total Revenue = 152560647.70
Jessore Total Quantity = 148899.0, Total Revenue = 150101620.87
Khulna Total Quantity = 150216.0, Total Revenue = 151179004.83
Mymensingh Total Quantity = 149714.0, Total Revenue = 149617733.40
Narayanganj Total Quantity = 149013.0, Total Revenue = 148949739.71
Noakhali Total Quantity = 148368.0, Total Revenue = 148340471.11
Pabna Total Quantity = 150484.0, Total Revenue = 150337583.51
Rajshahi Total Quantity = 150057.0, Total Revenue = 151279418.92
Rangpur Total Quantity = 150550.0, Total Revenue = 151273175.84
Sylhet Total Quantity = 148271.0, Total Revenue = 149407115.56
Tangail Total Quantity = 149128.0, Total Revenue = 149595760.61

C:\hadoop\sbin>hdfs dfs -cat /ecommerce/mr_output_payment/part-r-00000
Bank Transfer 149814
Card 149879
Cash 150059
Mobile Banking 150248
```

Figure 2: City-wise and payment-method-wise aggregation using Hadoop MapReduce

The results show that quantities and revenue are fairly evenly distributed across the twenty cities, each generating between approximately 148 and 153 thousand units and 148 million to 153 million in revenue. Among payment methods, Mobile Banking recorded the highest number of orders (150,248), closely followed by Cash (150,059).

8. Apache Pig Data Analysis

Apache Pig was used to perform higher-level analysis of the e-commerce sales dataset. The Pig script loaded the cleaned dataset from HDFS and performed grouping, filtering, summation, and average calculations to compute five separate analyses.

The following analyses were performed:

- Category-wise average unit price.
- Category-wise total quantity sold.
- Category-wise total revenue.
- City-wise average order revenue.
- Payment-method-wise total revenue.

The Pig script (`sajid_ecommerce_analytics.pig`) used to perform these operations is shown below:

```
sales = LOAD '/ecommerce/inputs/sales_data.csv' USING PigStorage(',')
AS (order_id:chararray, order_date:chararray, customer_id:chararray,
    customer_name:chararray, city:chararray, product_id:chararray,
```

```

        product_name:chararray, category:chararray, quantity:int,
        unit_price:double, revenue:double, payment_method:chararray);

clean_sales = FILTER sales BY order_id != 'order_id' AND quantity IS NOT NULL;

-- Category-wise average unit price
cat_group1 = GROUP clean_sales BY category;
category_avg_price = FOREACH cat_group1 GENERATE group AS category,
    AVG(clean_sales.unit_price) AS avg_price;

-- Category-wise total quantity
cat_group2 = GROUP clean_sales BY category;
category_quantity = FOREACH cat_group2 GENERATE group AS category,
    SUM(clean_sales.quantity) AS total_quantity;

-- Category-wise total revenue
cat_group3 = GROUP clean_sales BY category;
category_revenue = FOREACH cat_group3 GENERATE group AS category,
    SUM(clean_sales.revenue) AS total_revenue;

-- City-wise average order revenue
city_group1 = GROUP clean_sales BY city;
city_avg_revenue = FOREACH city_group1 GENERATE group AS city,
    AVG(clean_sales.revenue) AS avg_revenue;

-- Payment-method-wise total revenue
pay_group = GROUP clean_sales BY payment_method;
payment_revenue = FOREACH pay_group GENERATE group AS payment_method,
    SUM(clean_sales.revenue) AS total_revenue;

```

The script was executed using:

```
pig -x mapreduce sajid_ecommerce_analytics.pig
```

The resulting output directories in HDFS were:

- /ecommerce/pig_output/category_avg_price
- /ecommerce/pig_output/category_quantity
- /ecommerce/pig_output/category_revenue
- /ecommerce/pig_output/city_avg_revenue
- /ecommerce/pig_output/payment_revenue

The output of each analysis, viewed using `hdfs dfs -cat`, is shown in the figures below.

```

C:\big_data_project\sales_data\ecommerce_hadoop_project\pig>hdfs dfs -cat /ecommerce/pig_output/category_avg_price/part-r-00000
Toys,1002.0927884583311
Books,1010.4033408931826
Beauty,1002.1257454448255
Sports,1003.230413504486
Grocery,1003.7117876763173
Clothing,1005.3702719797477
Furniture,1002.967217780307
Automotive,1006.4333336676941
Electronics,1007.5722334889542
Home Appliances,1006.7754330787661

C:\big_data_project\sales_data\ecommerce_hadoop_project\pig>hdfs dfs -cat /ecommerce/pig_output/category_quantity/part-r-00000
Toys,298854
Books,299315
Beauty,299577
Sports,301912
Grocery,302956
Clothing,299924
Furniture,297836
Automotive,300353
Electronics,298550
Home Appliances,299811

C:\big_data_project\sales_data\ecommerce_hadoop_project\pig>hdfs dfs -cat /ecommerce/pig_output/category_revenue/part-r-00000
Toys,2.994573383600036E8
Books,3.0239574374999833E8
Beauty,2.9984364452999973E8
Sports,3.029172638599977E8
Grocery,3.037467996799992E8
Clothing,3.016383198199991E8
Furniture,2.983833019900004E8
Automotive,3.025033694099999E8
Electronics,3.0065963897999674E8
Home Appliances,3.0149702774999774E8

```

Figure 3: Category-wise average unit price, quantity, and revenue (Apache Pig output)

```
C:\big_data_project\sales_data\e-commerce_hadoop_project\pig>hdfs dfs -cat /e-commerce/pig_output/city_avg_revenue/part-r-00000
Feni,5010.211198230792
Bogra,5034.612822367567
Dhaka,5021.644538269134
Pabna,4984.006879392633
Khulna,5030.74788958774
Sylhet,5033.084573353559
Barisal,5036.265556288957
Comilla,5046.370551762807
Gazipur,5064.085763128217
Jessore,5038.1506014835595
Rangpur,4998.617977067732
Tangail,5017.071307191749
Dinajpur,5001.706214751259
Faridpur,5017.061837238796
Noakhali,4996.647504378874
Rajshahi,5044.665163398634
Chattogram,5019.609864179413
Coxs Bazar,5036.945783843229
Mymensingh,4990.085495113915
Narayanganj,5011.767823351309

C:\big_data_project\sales_data\e-commerce_hadoop_project\pig>hdfs dfs -cat /e-commerce/pig_output/payment_revenue/part-r-00000
Card,7.543319097899984E8,149879
Cash,7.541534764499822E8,150059
Bank Transfer,7.511488471400073E8,149814
Mobile Banking,7.534082147500014E8,150248
```

Figure 4: City-wise average revenue and payment-method-wise total revenue (Apache Pig output)

9. Results

The analysis produced the following major results, computed across all 600,000 transaction records:

Metric	Result
Total Records Processed	600,000
Total Quantity Sold	2,999,088 units
Total Revenue	₳ 3,013,042,448.13
Highest Revenue Category	Grocery (₳ 303,746,799.68)
Highest Revenue City	Comilla (₳ 153,011,001.50)
Highest Revenue Payment Method	Card (₳ 754,331,909.79)
Highest Order Count Payment Method	Mobile Banking (150,248 orders)

The dataset shows a remarkably balanced distribution of sales across product categories, cities, and payment methods, with Grocery generating the highest category revenue and Comilla generating the highest city-level revenue. Overall revenue across all 20 cities and all payment methods differs by only a few percentage points, indicating a well-distributed simulated transaction pattern.

10. Implementation Summary

The project successfully demonstrated the three required Big Data operations.

10.1 HDFS Storage Operation

HDFS was used to store the 600,000-record e-commerce dataset. The dataset was uploaded to the HDFS input directory and verified using the Hadoop NameNode web interface.

10.2 MapReduce Processing Operation

MapReduce was used to process the transaction records and calculate city-wise quantity and revenue as well as payment-method-wise order counts. The Mapper emitted key-value pairs for each transaction, while the Reducer aggregated the values for each group.

10.3 Pig Analysis Operation

Apache Pig was used to perform five analytical operations on the dataset — category-wise average price, category-wise quantity, category-wise revenue, city-wise average revenue, and payment-method-wise revenue — using grouping, filtering, summation, and average functions in Pig Latin.

11. Advantages

- Efficient distributed storage of a large 600,000-record dataset using HDFS.
- Scalable processing of city-wise and payment-wise statistics using MapReduce.
- Simple, readable analytical queries using Pig Latin instead of low-level Java code.
- Ability to process large transactional datasets that would be difficult to analyse manually.
- Clear separation of storage, processing, and analysis stages.
- A reproducible, command-line-based analytics workflow.

12. Limitations

- The project uses a single-machine (pseudo-distributed) Hadoop environment rather than a real multi-node cluster.
- The MapReduce implementation relies on straightforward CSV parsing without handling malformed records.
- Advanced visualization tools (dashboards, charts) were not used.
- Real-time/streaming data processing is not implemented.
- The dataset represents a fixed historical snapshot rather than live e-commerce activity.

13. Future Improvements

- Use Apache Spark for faster, in-memory large-scale processing.
- Use Hive for SQL-based Big Data analysis.
- Develop interactive dashboards using Power BI or Tableau.
- Implement product-level and customer-level sales analysis.
- Implement customer segmentation based on purchase behaviour.
- Perform monthly and yearly sales trend analysis.
- Develop predictive sales and demand forecasting models.
- Deploy the project on a real multi-node Hadoop cluster.

14. Conclusion

This project successfully demonstrated an end-to-end Big Data analytics workflow for e-commerce sales data using Hadoop, MapReduce, and Apache Pig. HDFS was used to store a 600,000-record transactional dataset, MapReduce was used to compute city-wise and payment-method-wise aggregates, and Apache Pig was used to perform category-wise, city-wise, and payment-wise analytical operations.

The analysis identified a total revenue of approximately ₹3,013,042,448.13 generated from 2,999,088 units sold, with Grocery as the highest-revenue product category and Comilla as the highest-revenue city. Overall, the project provided practical, hands-on experience with Hadoop-based Big Data storage, processing, and analysis, and demonstrated how these technologies can be applied to large-scale e-commerce transaction data.

15. References

1. Apache Software Foundation. Apache Hadoop Documentation.
2. Apache Software Foundation. Apache Pig Documentation.
3. White, T. Hadoop: The Definitive Guide.
4. Apache Software Foundation. Apache Hadoop MapReduce Documentation.